Nathan Gong

(609) 454-1896 | nathangong9@gmail.com | linkedin.com/in/nathangong9 | github.com/nathan-gong

EDUCATION

Columbia University | New York City, NY 2024 - 2026
Master of Science in Computer Science. GPA: 3.97/4.0
Coursework: High-Performance ML, AI, Computer Vision, Computational Robotics, NLP, HCI, Databases, Networks

Northeastern University | Boston, MA 2019 - 2023
Bachelor of Science in Bioengineering. Minor in Computer Science. GPA: 3.96/4.0
Honors: University Honors Program, President's Award, Tau Beta Pi, PEAK Research Award

SKILLS

Languages: Python, C++, Java, JavaScript, SQL, C, Bash
Libraries/Frameworks: PyTorch, Flask, Spring Boot, Pytest, GoogleTest, NumPy, Pandas, Scikit-learn, Matplotlib
Tools/Platforms: Git, Docker, GCP, Jira, Bamboo, Linux, QNX, Windows, MySQL, MongoDB, Neo4j, Jupyter
Practices: TDD, OOP, Agile, CI/CD, SDLC

EXPERIENCE

Software Engineer II | *Medtronic* | Boston, MA Jan 2022 - Present

- Develop production software in an embedded, mixed-OS (Linux, QNX, Windows) environment for a surgical robotics platform, the company's highest-funded product.
- Integrated a failsafe system to reduce error recovery time by 70% and enhanced reliability during clinical procedures, impacting hundreds of cross-functional stakeholders and accelerating U.S. regulatory clearance.
- Refactored the core GUI framework onto the new failsafe architecture and built an alarm routing tool to propagate messages across distributed compute nodes, enabling end-to-end testability and improving real-time error visibility.
- Migrated platform software to a unified Linux OS, implementing core C++ library enhancements to harden system security through least-privilege execution of services and application image authentication.
- Designed and validated manufacturing tooling supporting a production line transfer, and strengthened build scripts with structured error handling and CI-integrated unit test suites.
- Built a full-stack Python application to aggregate web data and generate risk reports for 30+ off-the-shelf software products, streamlining the user experience and improving report generation time by 80%.

Teaching Assistant | *Northeastern University* | Boston, MA Sep 2020 - Aug 2022

- Mentored 100+ students by providing technical guidance, debugging code, and explaining core algorithmic concepts.

Software Engineer Co-op | *iRhythm Technologies* | San Francisco, CA Jan 2021 - Jun 2021

- Automated firmware testing infrastructure for an ambulatory cardiac monitoring device, reducing test time by 50%
- Designed APIs to interface with manufacturing software tools and configure embedded systems for production.

PROJECTS

Roofline-Aware LLM Inference Analysis Jan 2026 - May 2026

- Optimized Llama-3.1-8B inference on an NVIDIA L4 GPU by implementing FlashAttention-2 and 4-bit quantization, achieving a 63% memory reduction at an 18% decode-throughput cost.

Robotic Tree Pruning Simulator Sep 2025 - Dec 2025

- Built a motion-planning pipeline to autonomously prune a simulated tree with a robotic arm that segments 3D point clouds to localize cut points, reconstructs surface meshes for collision modeling, and plans cutting trajectories.

Movie Data Visualization Application Nov 2024 - Dec 2024

- Wrote an ETL pipeline and web application to scrape and clean thousands of records from online sources, persist them across relational/document/graph stores, and visualize movie and actor popularity trends over time.